

UbuntuRemit: ASCO Implementation Approach

1. Implementation Strategy overview

The implementation of the Agentic Settlement and Compliance Orchestrator (ASCO) is divided into a rapid prototyping phase (to prove multi-agent viability) and a subsequent enterprise hardening phase. The focus is on leveraging the AMD ROCm ecosystem to achieve high-performance inference on private infrastructure.

2. Phase 1: Prototype and Hardware Validation

The immediate goal is to establish the base infrastructure and prove the core "Tug-of-War" negotiation concept between the agents.

- **Environment Setup:** Provision an AMD Instinct MI300X instance. Install the ROCm 7.0 stack and configure vLLM for optimized memory management and continuous batching.
- **Model Deployment:** Deploy two distinct open-weight models concurrently. A larger model (70B+ class) will be assigned the Compliance Sentinel role, while a math-optimized model will handle the Liquidity Strategist role.
- **Agent Handshake Design:** Implement a JSON-schema-based communication protocol to ensure agents can exchange findings deterministically without hallucinating payload structures.

3. Phase 2: Enterprise Hardening and Domain-Driven Design

Following the successful prototype, the system architecture must be refactored to meet the strict

standards of South African financial institutions and tech consultancies.

- **Domain Segmentation:** Apply Domain-Driven Design (DDD). The underlying logic for "Financial Routing" must be decoupled from the "LLM Inference" logic. This ensures that if AI models are swapped, the core banking business logic remains unaffected.
- **The Hybrid Guardrail Pattern:** Integrate hard-coded, rule-based gates at the entry and exit points of the agentic workflow. This guarantees that AI negotiation only occurs within legally permissible boundaries.
- **Audit Logging:** Develop a robust telemetry system that captures every prompt, response, and negotiation state, translating it into a format compliant with regulatory auditing standards.

4. Risk Management

Risk Vector	Impact	Mitigation Strategy
Non-Deterministic Output	Regulatory fines due to inconsistent AML application.	Implement the rule-based exit validator. The LLM suggests, but the final commit is checked against static logic.
High Latency	Failure to meet RTGS (Real-Time Gross Settlement) SLA windows.	Utilize FP8 or FP4 quantization techniques within the ROCm ecosystem to accelerate inference times without losing reasoning capability.

5. Next Steps

The immediate priority is to finalize the schema for the ISO 20022 pain.001 injection and establish the base connection to the MI300X instance via the developer portal.